

DPT

Probability - 3

- Two dice are thrown. What is the probability that the product of the numbers lie between 7 and 13
(a) $1/18$ (b) $1/6$ (c) $1/4$ (d) $1/2$
- A coin is tossed three times then find the chance of getting at least two heads :
(a) $1/2$ (b) $1/4$ (c) $1/8$ (d) None
- What is the chance of getting an odd number each time in two successive throws of a pair of dice :
(a) $1/2$ (b) $1/4$ (c) $3/4$ (d) None
- A has 3 shares in a lottery which has 3 prizes and 6 blanks and B has one share in another lottery which has 1 prize and 2 blanks. Ratio of their chance of winning is :
(a) 16:21 (b) 7:16 (c) 16:7 (d) 21:16
- A has 3 shares in a lottery containing 3 prizes and 9 blanks, B has two shares in another lottery containing 2 prizes and 6 blanks. Ratio of their chances of success is :
(a) 34:13 (b) 952:725
(c) 715:952 (d) 952:715
- A party of n persons sit at a round table then odds against two particular persons sit together will be :
(a) $n-1:2$ (b) $2:n-1$
(c) $n-3:2$ (d) $2:n-3$
- The first 12 letters of the alphabet are written at random what is the chance that there are 4 letters between A and B :
(a) $1/6$ (b) $7/11$ (c) $7/66$ (d) None
- An urn contains 10 white and 3 blank balls another urn contains 3 white and 5 blank balls. Two balls are transferred from the first urn placed in the second urn then one ball is taken from the later, what is the probability that it is a white ball :
(a) $2/13$ (b) $3/25$
(c) $15/52$ (d) $177/130$
- One urn contains 3 white and 2 black balls, another contains 5 white and 3 black balls. If an urn is chosen at random and a ball is taken from it what is the chance that it is white :
(a) $3/10$ (b) $5/16$ (c) $3/8$ (d) $49/80$
- If A and B are two events $P(A \cup B) = 7/8$, $P(A \cap B) = 1/4$ then $P(A) =$
(a) $3/8$ (b) $1/4$ (c) $3/4$ (d) None
- If $P(A) = .4$, $P(B) = x$ and $P(A \cap B) = .2$ and A and B are independents, then $x =$
(a) $1/3$ (b) $1/2$ (c) $2/3$ (d) None
- $P(E_1/E_2) = \dots\dots\dots$ (where E_1 and E_2 are two events)
(a) $P(E_1) + P(E_2)$ (b) $P(E_1) - P(E_2)$
(c) $P(E_1 \cap E_2)/P(E_2)$ (d) $P(E_1 \cap E_2)/P(E_1)$
- If A and B are independent event then $P(A \cap B) =$
(a) $P(A) + P(B)$ (b) $P(A) \cdot P(B)$
(c) $P(A/B)$ (d) $P(B/A)$
- If A and B are two independent events. The Probability that both A and B occur is $1/6$ and the probability of that neither of them occurs is $1/3$. The probability of the occurrence of A is :
(a) $1/2$ (b) $1/5$ (c) $1/4$ (d) None
- 15 coupons are numbered 1,2,3,.....,14,15, seven coupons are selected at random, one at a time with replacement. The probability that 9 would be the largest number appearing on the selected coupon is :
(a) $(1/16)^6$ (b) $(8/15)^7$
(c) $(3/5)^7$ (d) None
- A class consists of 80 students 25 of them are girls and 55 boys. If 10 of them are rich and the remaining poor and also 20 of them are intelligent then the probability of selecting an intelligent rich girl is :
(a) $5/128$ (b) $25/128$
(c) $5/64$ (d) None
- A bag contains 8 black and 5 white balls. Two balls are drawn at random. Of which of the following events the probability is maximum :
(a) Both the balls are white
(b) One ball is white and another is black
(c) Both the balls are black
(d) Each of the above events has the same probability
- Four coins are tossed once. The probability of not getting at least one tail is :
(a) $15/16$ (b) $1/16$ (c) $4/5$ (d) $1/4$
- In a sample survey of a village there are 1500 boys and 1475 girls then probability that a baby born is a boy :
(a) $1475/1500$ (b) $1500/1475$
(c) $60/119$ (d) $300/695$

20. Four tickets marked 00, 01, 10, 11 respectively are placed in a bag. A ticket is drawn at random five times, being replaced each time. The probability that the sum of the numbers on tickets thus drawn is 23 is :
(a) $25/256$ (b) $2/256$
(c) $1/256$ (d) None
21. A speaks the truth 2 times, out of 3 and B, 4 times out of 5 ; they agree in assertion that from a bag containing 6 balls of different colours a red ball has been drawn. The probability that the statement is true is :
(a) $40/41$ (b) $3/41$ (c) $1/41$ (d) None
22. A bag contains 3 white and 3 red balls and a second bag contains 4 white and 5 red balls. From the two bags, a ball is drawn at random and it is a red ball. The probability that it is taken from second bag is :
(a) $25/52$ (b) $1/26$
(c) $27/52$ (d) None
23. Four dice are thrown. The probability that the sum of numbers appearing on the dice is 18 is :
(a) $5/81$ (b) $1/81$ (c) $2/81$ (d) None
24. Two dice are thrown simultaneously. The probability that their sum is a multiple of 4 or 6 is :
(a) $1/9$ (b) $13/36$ (c) $7/14$ (d) $5/9$
25. Twenty tickets are marked by first 20 natural numbers. The probability that a ticket taken at random shows a number which is a multiple of 3 or 7 is :
(a) $9/20$ (b) $7/20$ (c) $2/5$ (d) None
26. Three dice are thrown. The probability that they never show same number is :
(a) $1/6$ (b) $1/18$ (c) $35/36$ (d) $3/28$
27. When an experiment is performed once , the probability of an event A occurring is 0.4 . If the same experiment is performed independently 3 times, then the probability of A occurring atleast once is :
(a) 0.936 (b) 0.784 (c) 0.904 (d) None
28. A tosses 3 coins and B tosses 2 coins. The probability that A gets more heads than B is :
(a) $1/3$ (b) $2/3$ (c) $1/2$ (d) None
29. The chance of happening of an event is the square of happening of the other. But the odds against the first event is cube of odds against second event. The probabilities of the two events are :
(a) $1/9$ and $1/3$ (b) $2/9$ and $2/3$
(c) $1/3$ and $1/9$ (d) None of these
30. From a pack of 52 cards the cards are drawn till an ace appears. The probability that an ace does not come in first 26 cards is :
(a) $46/153$ (b) $23/27$
(c) $109/153$ (d) None

SCORE SHEET

PROBABILITY - 03

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| 1. (c) | 2. (a) | 3. (b) | 4. (c) | 5. (d) | 6. (c) | 7. (c) | 8. (d) | 9. (d) | 10. (d) |
| 11. (b) | 12. (c) | 13. (b) | 14. (a) | 15. (c) | 16. (a) | 17. (c) | 18. (b) | 19. (c) | 20. (a) |
| 21. (d) | 22. (d) | 23. (d) | 24. (b) | 25. (c) | 26. (a) | 27. (b) | 28. (c) | 29. (a) | 30. (d) |

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